Syllabus

UNIT 1: Indian Mathematics & Foundations of ML (6 Hours)

- Historical survey of Indian Mathematics: number theory, combinatorics, geometry.
- Algorithmic thinking in ancient mathematical texts.
- Overview of Machine Learning: supervised vs. unsupervised learning.
- How mathematical thinking forms the foundation for ML.

Unit Outcomes (UO1):

- Understand contributions from Indian Mathematics and map them to modern computational ideas.
- Explain the role of mathematics in ML model building.
- Identify areas where traditional combinatorial and geometric reasoning support ML tasks.

UNIT 2: Indian Mathematics – Number Theory, Combinatorics & Algorithms

- Modular arithmetic, residues, divisibility (Indian number theory tradition).
- Combinatorial counting methods → applications in ML (search, optimization).
- Approximation methods, infinite series, and early Indian trigonometric/geometry approaches.
- Linking Indian algorithms with modern algorithmic principles.

Unit Outcomes (UO2):

- Apply combinatorial and number-theoretic reasoning to algorithmic problems.
- Explain the link between traditional approximation methods and iterative ML solvers.
- Understand how pattern-based mathematical thinking translates into algorithm design.

UNIT 3: Linear Algebra for Machine Learning

- Vectors, matrices, linear transformations, systems of equations.
- Rank, subspaces, orthogonality.
- Eigenvalues, eigenvectors, diagonalization.
- Singular Value Decomposition (SVD).
- Principal Component Analysis (PCA).

Unit Outcomes (UO3):

- Perform matrix operations and understand geometric interpretations.
- Apply eigenvalues/eigenvectors and SVD in dimensionality reduction.
- Relate linear algebra tools directly to ML models such as PCA and regression.

UNIT 4: Probability, Statistics & Optimization for ML

- Probability axioms, Bayes' theorem.
- Random variables and distributions (normal, binomial).
- Joint, marginal, and conditional distributions.
- Covariance, correlation matrices.
- Central Limit Theorem & LLN.
- Gradients, Hessians, gradient descent.
- Constrained optimization (Lagrange multipliers).

Unit Outcomes (UO4):

- Apply probability and statistical concepts to model uncertainty in ML.
- Use covariance and correlation to analyse dataset structure.
- Understand optimization strategies behind ML algorithms such as gradient descent.

UNIT 5: ML Algorithms & Indian Mathematical Perspective

- Supervised learning: Linear Regression, Logistic Regression.
- Unsupervised learning: k-Means, PCA interpretation.
- Indian mathematics insights for pattern recognition and combinatorial reasoning.
- Approximation methods → gradient-based ML methods.
- Case studies, datasets, and mini-project ideas.

Unit Outcomes (UO5):

- Implement basic ML algorithms with mathematical understanding.
- Connect Indian mathematical thinking to modern ML concepts like clustering and pattern recognition.
- Analyse case studies where Indian knowledge principles inspire modern computational models.

4. Pedagogy

- Lectures + hands-on problem solving
- ML demonstrations using Python
- Flipped classroom
- Case studies linking IKS and ML
- Mini–project at the end

5. Assessment Scheme

- Internal Assessment / Assignments – 30M
- Final Examination – 70M

6. Mapping with IKS

This course integrates IKS by:

- Introducing classical Indian mathematical techniques and concepts.
- Linking Indian combinatorics, geometry, series, and algorithmic methods to ML theory.
- Demonstrating how India's historical mathematical advances form part of the conceptual foundation of modern computational thinking.
- Encouraging students to critically compare traditional and modern approaches.